

AI Guard

Host-Based Intrusion Detection System

Detection Rule Documentation

Python-based defensive cybersecurity internship project

This document describes the custom detection rules, alert severity classification, storage mechanism, AI-assisted analysis and testing methodology used in the AI Guard HIDS prototype.

1. Project Overview

AI Guard is a Python-based Host-Based Intrusion Detection System designed to monitor Linux security logs in real time. The prototype uses custom detection rules to identify suspicious authentication and privilege-related activity.

Detected events are stored in an SQLite database and displayed through a Streamlit dashboard. The prototype also integrates the Groq API to provide AI-assisted security event explanations and remediation guidance.

2. Detection Architecture

The monitoring workflow consists of the following stages:

- Linux security log or simulated security log generates an event.
- Watchdog monitors the log file for newly appended entries.
- The detector analyzes each new log line using custom rules.
- Detected events are stored in SQLite.
- The Streamlit dashboard displays current and historical alerts.
- Groq AI can analyze selected events and provide remediation guidance.

3. Failed Login Detection

Objective: Detect unsuccessful SSH authentication attempts recorded in the monitored Linux authentication log.

Detection pattern: The rule searches for failed password messages containing a source IP address.

Example: Failed password for invalid user admin from 192.168.56.101

Severity: LOW

Response: The detected event is stored in SQLite with timestamp, source IP, severity and the original log message.

4. Brute Force Detection

Objective: Detect repeated failed authentication attempts from the same source IP address.

Detection logic: Five or more failed login attempts from the same IP address within a 60-second window are classified as a brute-force attempt.

Parameter	Value
Threshold	5 failed attempts
Time Window	60 seconds
Severity	HIGH

Response: The event is stored in SQLite and can be analyzed through the Groq-powered AI security assistant.

5. Privilege Escalation Detection

Objective: Identify suspicious sudo events involving root-level privileges.

Detection pattern: The rule searches for sudo log entries containing USER=root.

Example: sudo: kali : user NOT in sudoers ; USER=root ; COMMAND=/bin/bash

Severity: HIGH

Response: The event is stored as a privilege escalation alert and displayed on the dashboard.

6. Port Scan Detection

Objective: Identify port-scanning activity represented in the monitored security log.

Detection pattern: The system detects controlled log entries containing PORT_SCAN followed by a source IP address.

Example: PORT_SCAN detected from 127.0.0.1

Severity: MEDIUM

Validation: Port scanning functionality is validated in an authorized local laboratory using Nmap. A controlled port-scan event is also used to test the HIDS detection rule.

7. Alert Severity Classification

Severity	Meaning
LOW	Individual suspicious event requiring review
MEDIUM	Suspicious network activity requiring investigation
HIGH	Activity that may indicate an active attack or privilege-related threat

8. Alert Storage

Detected security events are stored in an SQLite database.

- Alert ID
- Timestamp
- Event Type
- Source
- Severity
- Original Log Message
- Status

9. AI-Assisted Security Analysis

The prototype integrates the Groq API to provide AI-assisted analysis of detected security events.

- Explanation of what happened
- Why the event may be a security concern
- Recommended remediation steps

- Suggested administrator checks

The AI assistant is intended to support security analysis and does not replace human investigation or professional security operations.

10. Testing Methodology

- Failed login events were generated in the controlled log file.
- Five failed login attempts from the same source were used to validate brute-force detection.
- A controlled sudo USER=root event was used to validate privilege escalation detection.
- Nmap was used against an authorized local laboratory target for port-scanning validation.
- The Streamlit dashboard was used to verify stored alerts.
- Groq AI analysis was tested using detected security events.

11. Limitations

This internship prototype uses simulated Linux log data for demonstration and controlled testing. It is not intended to represent complete enterprise-grade intrusion detection coverage.

The current prototype focuses on selected authentication, privilege and port-scan detection rules. Additional detection rules, endpoint telemetry and machine-learning capabilities can be added in future development phases.